UNIT 3 · TOPIC 10(B)

Estimating Energy and Power Requirements

Syllabus topic: Matching power-implement system; estimation of energy and power requirements

Energy per hectare

$$\text{E (kWh/ha)} = \text{power (kW)} \times \text{field time (h/ha)} \text{ — the practical route}$$

Example. 21 kW drawbar at EFC 0.7 ha/h → **30 kWh/ha** for primary tillage.

$$\text{E (kWh/ha)} = \text{unit draft (kPa)} \times \text{depth (m)} \times 10^4 \div 3600$$

Corrected formula. The source notes divide by 3.6×10^6 here, which holds only when unit draft is in Pa (N/m^2); with kPa the divisor is 3600. Check: 70 kPa × 0.15 m × 10^4 ÷ 3600 ≈ 29 kWh/ha — consistent with the 30 kWh/ha example and the MB-ploughing range below.

Operation	Typical energy requirement
Mouldboard ploughing	25–40 kWh/ha
Secondary tillage	8–15 kWh/ha
Sowing	3–6 kWh/ha

Links to Unit 7 energy audits.

Practice MCQs

Q132. Typical energy requirement of MB ploughing is about:

(a) 3–5
(b) 10–15
(c) 25–40
(d) 100–150 kWh/ha

Answers

Q132 — (c) 25–40

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 2 (Field Capacities and Cost Analysis); Hunt — Farm Power and Machinery Management; Jagdishwar Sahay; SMAM/CHC guidelines; NPTEL lectures.